

Continuous Random Variable

A continuous random variable is a variable that can take on any numerical value within a given range or interval, possessing an infinite number of decimal places and being uncountably infinite, unlike a discrete variable. It is often a measurement, with common examples including height, weight, temperature, and time.

Probability Density Function (PDF)

The probability density function defines the probability function representing the density of a continuous random variable lying between a specific range of values. In other words, the PDF produces the likelihood of values of the c.r.v.

$$P(a < X < b) = \int_a^b f(x)dx$$

Properties:

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$

Note:

1. $P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$
2. $P(X = x) = 0$

Practice Question

1. Let X be a continuous random variable with *pdf*:

$$f(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Verify that this is a valid *pdf*.
- (b) Find $P(0.25 < X < 0.75)$.

2. The lifetime (in 1000 hours) of a light bulb is modeled by a r.v. X with *pdf*:

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Find $P(1 < X < 3)$.

3. Let X be a continuous r.v. with the *pdf* given by:

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & x \geq 2. \end{cases}$$

Find $P(0.5 < X < 1.5)$.

4. The amount of bread (in hundreds of pounds) x that a certain bakery is able to sell in a day is found to be a numerical valued random phenomenon, with a probability function. Specified by the PDF $f(x)$, given by:

$$f(x) = \begin{cases} kx, & 0 \leq x < 5, \\ k(10 - x), & 5 \leq x < 10, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the value of k .
- (b) What is the probability that the number of pounds of bread that will be sold tomorrow is:
- more than 500 pounds,
 - less than 500 pounds, and
 - between 250 and 750 pounds.
- (c) Denoting by A , B , and C the events that the pounds of bread sold are as in $b(i)$, $b(ii)$, and $b(iii)$, respectively. Find $P(A|B)$, $P(A|C)$.
- Are A and B independent events?
 - Are A and C independent events?
5. Let X be a continuous r.v. with *p.d.f.* given by:

$$f(x) = \begin{cases} kx, & 0 \leq x < 1, \\ k, & 1 \leq x < 2, \\ -kx + 3k, & 2 \leq x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Determine the constant k ,
- (b) Determine $F(x)$, the *cdf*, and
- (c) If x_1, x_2, x_3 are three independent observations from X , what is the probability that exactly one of these three numbers is larger than 1.5?.

6. Suppose that the time (in minutes) that a person has to wait at a certain station for a train is a random phenomenon, specified by the distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ \frac{x}{8}, & 0 \leq x < 2, \\ \frac{x^2}{16}, & 2 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

- (a) Is the distribution function continuous? If so, give the formula for its probability density function.
- (b) What is the probability that a person will have to wait
 - i. more than 2 minutes,
 - ii. less than 2 minutes, and
 - iii. between 1 and 2 minutes?
- (c) What is the conditional probability that the person will have to wait for a train
 - i. more than 2 minutes, given that it is more than 1 minute,
 - ii. less than 2 minutes, given that it is more than 1 minute?